

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Midsemester Exam #2

Only one sheet of paper 8 1/2 x 11 is allowed

No phones or i-pads or computers are allowed

Duration: 70 minutes --- October 21, 2025

All Problems must be done

Problem 1: (50 points)

Let's design a shell and tube heat exchanger to heat a stream with $m_c C_{p,c} = 5,000 \text{ BTU} / \text{h}^\circ\text{F}$ from 90°F to 210°F flowing in the tubes using as the heat source a stream with $m_h C_{p,h} = 10,000 \text{ BTU} / \text{h}^\circ\text{F}$ entering the shell at 260°F . The heat exchanger will contain tubes with a 0.5 in. outside diameter. In order to maintain a reasonable fluid velocity inside the tubes, $N_t = 30$ tubes per pass will be used. The overall heat transfer coefficient is $U_o = 150 \text{ BTU} / \text{h ft}^2 \text{ }^\circ\text{F}$. For a heat exchanger with one-tube pass and one-shell pass, determine the required tube length. **(40 points)**

Suppose that we need to have shorter tubes, what can we do? **(10 points)**

I did not have this formula down in the equationsheet. 10?

If we have shorter tube, we can increase number of number of tube (likely unfeasible) 10

- or.
- Increase inlet heating stream temperature
 - reduce stream velocity (by lowering inlet pressure)
 - If current set up as concurrent flow, can switch to counter flow instead (may not be feasible)
 - increase radius of tube (maybe unfeasible)



$$N_t = 30$$

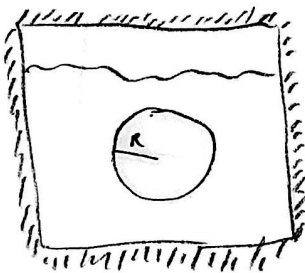
$$m c \Delta T_{Al} = m c \Delta T_w$$

Problem 2: (60 points)

Let's consider a case where 1 kg of aluminum spheres with radius $R = 10$ mm initially at 90°C are immersed in 1 L of water initially at 20°C in a tank. The tank is insulated and heat losses are negligible. The water is assumed to be perfectly mixed. Determine the time needed to cool the aluminum spheres to 40°C assuming that the external heat transfer resistance at the sphere-fluid interface is dominant and that $h = 500 \text{ W/m}^2\text{K}$. The density of aluminum is $\rho = 2,700 \text{ kg/m}^3$ and its heat capacity is $C_p = 900 \text{ J/kg K}$. Assume that the water temperature remains constant. (30 points)

If the changing temperature of the water is considered, determine the time needed to cool the aluminum spheres to 40°C . (30 points)

a)



$$\rho_{Al} = 2700 \text{ kg/m}^3$$

$$C_p = 900 \text{ J/kg K}$$

$$T_i = 90^\circ\text{C} = 363 \text{ K}$$

$$T_f = 40^\circ\text{C} = 313 \text{ K}$$

$$h = 500 \text{ W/m}^2\text{K}$$

$$T_\infty = 20^\circ\text{C} = 293 \text{ K}$$

$$V_w = 1 \text{ L} = 10^{-3} \text{ m}^3$$

$$R = 10 \text{ mm} = 10^{-2} \text{ m}$$

$$C_{pw} = 4180 \frac{\text{J}}{\text{kg K}}$$

$$\rho_w = 1000 \text{ kg/m}^3$$

external heat transfer resistance at sphere-fluid interface is dominant $\Rightarrow$ Can use lump model

$$\Rightarrow \rho V_{Al} C_{pAl} \frac{dT}{dt} = -h A_s (T - T_\infty) \quad \therefore \quad \frac{dT}{T - T_\infty} = \frac{-h 4\pi R^2}{\rho \frac{4}{3}\pi R^3 C_{pAl}} dt$$

$$\text{I.C.: } T(t=0) = 363 \text{ K}$$

$$\Rightarrow \int_{T_i}^{T_f} \frac{1}{T - T_\infty} dT = \frac{-3h}{\rho R C_{pAl}} \int_0^{t_f} dt$$

$$\Rightarrow \ln |T - T_\infty| \Big|_{T_i}^{T_f} = \frac{-3h}{\rho R C_{pAl}} t_f$$

$$\Rightarrow \ln \frac{T_f - T_\infty}{T_i - T_\infty} = \frac{-3h}{\rho R C_{pAl}} t_f \quad \Rightarrow t_f = \frac{-\rho R C_{pAl}}{3h} \ln \frac{T_f - T_\infty}{T_i - T_\infty}$$

$$t_f = \frac{-2700 \cdot 10^{-2} \cdot 900}{3 \cdot 500} \ln \frac{313 - 293}{363 - 293}$$

$$t_f = 20.29 \text{ s}$$

30 + 5

$$T_{\infty} = T_w, T_w = T_{\infty}(t)$$

b) Water temperature does not stay constant.

Heat rate between Al and water is constant $\rightarrow m_{Al} C_{pAl} \frac{dT_{Al}}{dt} = m_w C_{pw} \frac{dT_w}{dt}$

$$\frac{dT_w}{dt} = \frac{m_{Al} C_{pAl}}{m_w C_{pw}} \frac{dT}{dt} \quad \leftarrow T_w \equiv T_{Al}(t)$$

$$\int_{T_{\infty,i}}^{T_{\infty}} \frac{dT_w}{dt} dt = \frac{\rho_{Al} V_{Al} C_{pAl}}{\rho_w V_w C_{pw}} \int_{T_i}^T \frac{dT}{dt} dt$$

$$T_{\infty} - 293 = \frac{2700 \cdot \frac{4}{3} \pi (10^{-2})^3 \cdot 900}{1000 \cdot 10^{-3} \cdot 4180} (T - 363)$$

$$T_{\infty} = 0.00244 (T - 363) + 293 \Rightarrow T_{\infty} = 0.00244 T + 292.116$$

$$\rho_{Al} V_{Al} C_{pAl} \frac{dT}{dt} = -h A_{s,Al} (T - T_{\infty}) \quad \text{I.C.: } T(t=0) = 363 \text{ K}$$

$$\frac{dT}{dt} = \frac{h 4\pi R^2}{\rho_{Al} \frac{4}{3} \pi R^3 C_{pAl}} (T - 0.00244 T - 292.116)$$

$$\frac{dT}{dt} = \frac{-3h}{\rho_{Al} R C_{pAl}} (0.998 T - 292.116)$$

$$\frac{dT}{dt} = \frac{-3 \times 500}{2700 \cdot 10^{-3} \cdot 900} (0.998 T - 292.116) = -\frac{5}{81} (0.998 T - 292.116)$$

$$\frac{dT}{dt} = -\frac{0.0616 T}{167} + \frac{18.032}{167}$$

$$\frac{dT}{-0.0616 T + 18.032} = dt \Rightarrow \int_{T_i}^{T_f} \frac{dT}{-0.0616 T + 18.032} = \int_0^{t_f} dt$$

$$\Rightarrow \frac{-1}{0.0616} \ln |18.032 - 0.0616 T| \Big|_{363}^{313} = t_f = 20.1 \text{ s}$$

✓ This should have been longer than in previous part, maybe rounding error?